

Supplementary Material: RotateAttention: RoPE-Aware Rotation and Range Rectification for INT4 Quantized Attention in Video Generation

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1 Theoretical Proof of RotateAttention Compatibility with SageAttention

Theorem 1. *Let $R \in \mathbb{O}(D)$ be an orthogonal matrix, and let the query tensor $Q \in \mathbb{R}^{N \times D}$ and key tensor $K \in \mathbb{R}^{N \times D}$ be smooth tensors. Define $\bar{Q} = \mathbb{E}_0[Q]$ and $\bar{K} = \mathbb{E}_0[K]$ as the sequence-wise means. Let $\hat{Q} = QR$ and $\hat{K} = KR$. The centered inner product remains invariant under the orthogonal transformation R :*

$$\begin{aligned} (Q - \bar{Q})(K - \bar{K})^T &= ((Q - \bar{Q})R)((K - \bar{K})R)^T \\ &= (QR - \mathbb{E}_0[QR])(KR - \mathbb{E}_0[KR])^T \end{aligned} \quad (1)$$

$$= (\hat{Q} - \mathbb{E}_0[\hat{Q}])(\hat{K} - \mathbb{E}_0[\hat{K}])^T. \quad (2)$$

According to Theorem 1, the smoothing operation and the application of the rotation matrix R are commutative and compatible. To maximize computational efficiency, we apply the rotation matrix R immediately following the RoPE encoding.

2 Compatibility with PAROAttention and VBench Evaluation

Because PAROAttention operates at the token level while the rotation and range-optimized P methods are applied at the channel level, the two approaches are orthogonal and can be seamlessly combined. We integrated our recommended half-rotation and range-optimized P methods into both SageAttention and PAROAttention to evaluate their combined impact.

As demonstrated in Tab. 1, combining our method with PAROAttention further enhances video quality across multiple models and metrics, achieving VBench scores that closely approach those of FP16 attention.

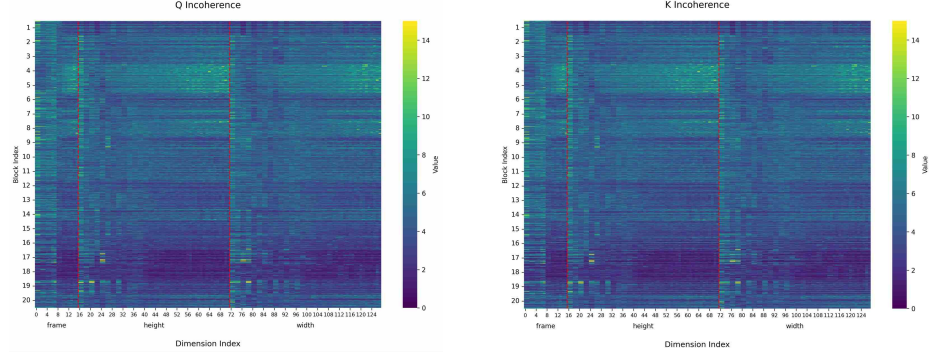
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Table 1: Comparison of Difference Metrics and Performance Compatible with PAROAttention (Wan2.2 and HunyuanVideo)

Model	Method			Difference Metrics				VBench $\uparrow$
	Datatype	PARO	Rotate	Cosine $\uparrow$	MSE $\downarrow$	SSIM $\uparrow$	PSNR $\uparrow$	
Wan2.2 -I2V	FP16	—	—	1.0000	0.0000	1.0000	∞	0.7842
	INT4	✓	×	0.9812	568.24	0.7612	22.214	0.7715
	INT4	✓	✓	0.9832	487.42	0.7853	23.086	0.7768
Wan2.2 -T2V	FP16	—	—	1.0000	0.0000	1.0000	∞	0.7625
	INT4	✓	×	0.9584	1395.2	0.6654	18.572	0.7492
	INT4	✓	✓	0.9592	1378.5	0.6702	18.694	0.7631
Hyvideo -T2V	FP16	—	—	1.0000	0.0000	1.0000	∞	0.8054
	INT4	✓	×	0.9735	1074.8	0.6634	18.982	0.7912
	INT4	✓	✓	0.9823	1028.6	0.6782	19.145	0.7985

3 Outlier Distribution in Another Model

**Fig. 1:** HunyuanVideo **Q/K** outlier distributions.

We provide the outlier distributions of **Q** and **K** in HunyuanVideo in Fig. 1. The results exhibit the same two patterns summarized in Sec.3 : **Spatiotemporal Segmental Incoherence** and **Cross-Matrix Symmetry**, suggesting that the observed **Q** and **K** outlier pattern is not specific to a single model.

4 4-step Distilled Wan2.2 with Different INT4 Ratio

We add experiments on a 4-step distilled Wan2.2-T2V from Wan-Lightning. The quantitative results are shown in Table 2, with visual examples in Fig. 2. Even at 100% INT4 attention, VBench remains close to FP16, while similarity to the FP16 output decreases as expected.

Notably, if the objective is to preserve output-level consistency between quantized-attention videos and full-precision FP16 videos, our current method

still has room for improvement on the 4-step distilled model. As shown in Table 2, quantizing only 25% of the attention layers to INT4 already reduces the cosine similarity to the FP16 output to 0.9358, suggesting that few-step distilled video generation models are particularly sensitive to attention quantization.

Table 2: 4-step distilled Wan2.2-T2V with different INT4 ratios.

INT4 Attn.	Similarity to FP16				VBench $\uparrow$
	Cos. $\uparrow$	MSE $\downarrow$	SSIM $\uparrow$	PSNR $\uparrow$	
0%	1.0000	0.0	1.0000	∞	0.765
25%	0.9358	1341.1	0.5407	16.94	0.761
50%	0.9152	1781.3	0.4487	15.68	0.758
63%	0.8957	2180.0	0.4173	14.80	0.759
100%	0.8772	2547.9	0.4152	14.19	0.753

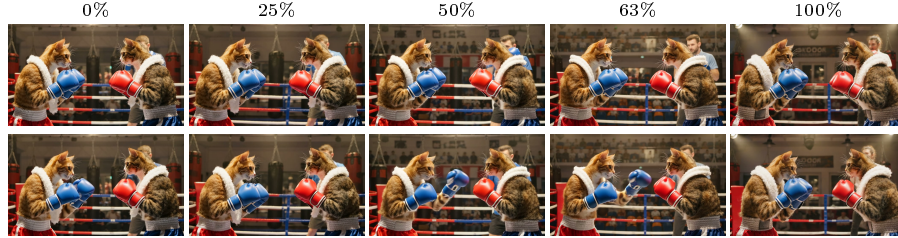


Fig. 2: First/end frames of 4-step distilled Wan2.2-T2V under different INT4 attention ratios.

5 Implementation Details

The interleaved rotation follows the RoPE layout and can be fused into the RoPE kernel. For half rotation, we split $\mathbf{Q}$ and $\mathbf{K}$ into three channel segments and apply efficient element-wise multiplications. $\mathbf{Q}$ and $\mathbf{K}$ use per-token scaling, $\mathbf{V}$ uses per-group channel scaling with group size 64, and $\mathbf{P}$ uses per-block attention scaling with a fixed scale.